KLH

SM Courier Service - Courier Delivery Route and Tracking System

COLLEGE EXPO PROJECT • CONSOLE-BASED JAVA APPLICATION

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Abstract

Project overview, key features and technologies

- Console-based courier management system built in Java.
- Customers create a courier, receive a unique ID, view charges and track packages.
- Admins manage statuses, revenue, summaries and charge-based sorting.
- Text-file storage keeps records available after restart.
- Built with OOP, exception handling, file I/O, collections and streams.

Understanding Problem Statement

From scattered manual records to one reliable workflow

Current challenges

- Courier details are easy to lose or mix up.
- Customers cannot easily check package status.
- Manual charge and revenue calculations cause mistakes.
- Data disappears when a basic program closes.

Proposed solution

A menu-driven system with separate Customer and Admin roles, automatic charge calculation, status tracking and persistent saved records.

CO 1 Topics

Java Fundamentals

Concept	Used in the project
Variables and data types	Storing weight, distance, charge and status
Operators	charge = 50 + (weight × 10)
Scanner input	Reading menu choices and courier details

CO 2 Topics

Control Statements

Concept	Used in the project
if / else	Delivery eligibility check (distance up to 500 km)
switch	Main menu, customer menu, admin menu and status selection
while loops	Menus repeat until the user chooses to go back or exit

CO 3 Topics

Methods

Concept	Used in the project
Methods with parameters and return values	DeliveryCharge(), CourierTracking(), DeliveryEligibility()
Static methods	Operations such as CreateDelivery(), UpdateStatus(), TotalIncome()
Modular design	Menus and operations split into separate methods

CO 4

Object-Oriented Design and Exception Handling

Concept	Used in the project
Classes and objects	Courier class; one object per courier
Constructors (overloading)	One for new couriers; one for loaded or sample couriers
Encapsulation	Private fields with getters and a setter
Static member	nextId counter generates unique IDs
toString()	Prints complete courier details
Exception handling	try-catch for invalid numbers; multi-catch while reading the file

CO 5

Strings, Sorting and File I/O

Concept	Used in the project
String methods	trim(), replace(), toUpperCase(), isEmpty() clean user input
split()	Reads each saved line from the file
Comparator	Sorts couriers by highest charge
File I/O	Files, Path, BufferedReader and BufferedWriter save and load couriers.txt
try-with-resources	File closes automatically

CO 6

Collections, Lambdas and Streams

Concept	Used in the project
ArrayList<Courier>	Stores all couriers, replacing the old arrays
Map	Holds the status counts
Lambda and method reference	findCourier(), Courier::getCharge
Stream filter + findFirst	Finds a courier by ID
Stream mapToInt + sum	Calculates total revenue
Stream groupingBy + counting	Counts couriers in each status

Conclusion

A practical application of all six course outcomes

The project turns a manual courier process into a working, menu-driven Java application.

It combines all six course outcomes, saves data between runs and handles invalid input safely.

Future scope

MySQL database

Route management and route map

Delivery time estimation

Customer and admin login

GUI

Real-time GPS tracking

Screen Shots

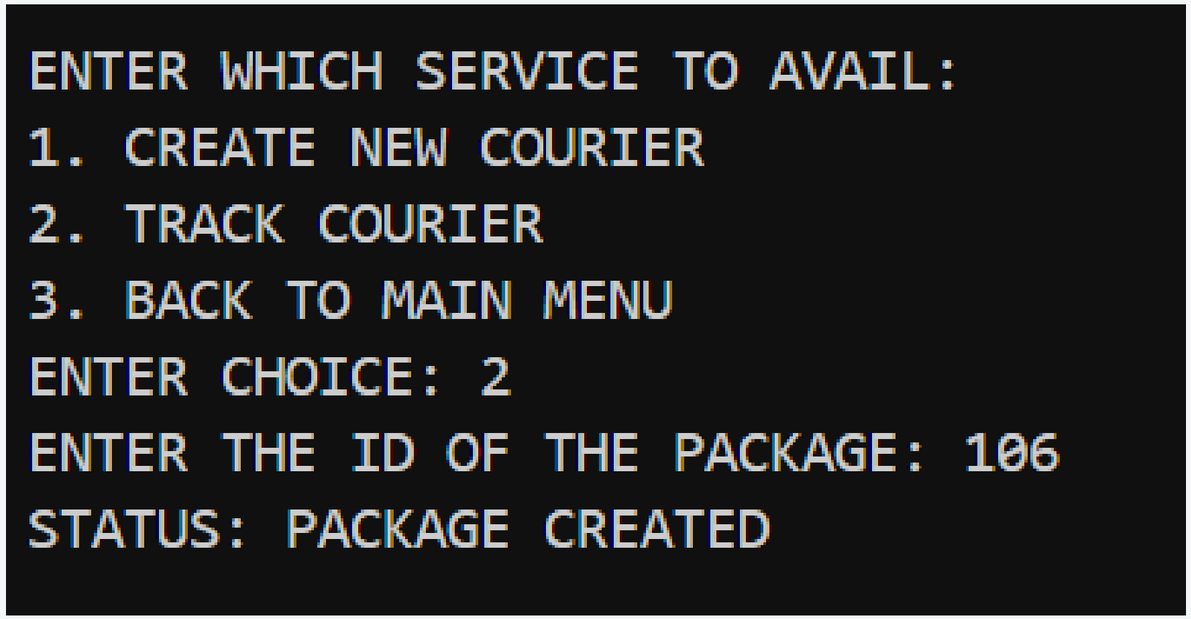
Application output

```
WELCOME TO SM COURIER SERVICE
CHOOSE YOUR ROLE
-----------------
1. CUSTOMER
2. ADMIN
3. EXIT PROGRAM
-----------------
ENTER CHOICE:
```

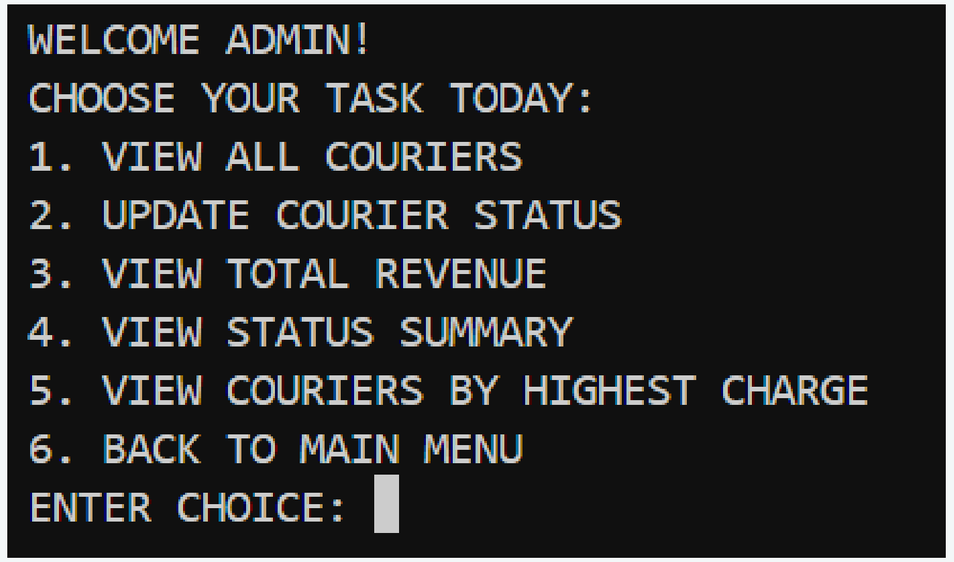
Main Menu

```
ENTER WHICH SERVICE TO AVAIL:
1. CREATE NEW COURIER
2. TRACK COURIER
3. BACK TO MAIN MENU
ENTER CHOICE: 1
ENTER NAME: sri
ENTER DESTINATION: hyderabad
ENTER WEIGHT OF THE PACKAGE: 56
ENTER THE DISTANCE: 300
YOUR PACKAGE ID: 106
COST: 610
```

Create Courier



Track Courier



Admin Menu

Geo Tag Photo and GitHub Link

Insert geo-tagged photo here

GitHub Repository Link

https://github.com/kitynoir09/Courier-Delivery-Route-and-Tracking